

LAB ASSIGNMENT -1

1. Write a program (WAP) to display "Hello World" on console display.

Code:

```
#include<iostream>
using namespace std;

int main()
{
    cout << "Hello World";
}
```

2. WAP to input an integer, decimal and character and display it.

Code:

```
#include<iostream>
using namespace std;

int main()
{
    int a;
    float b;
    char c;

    cin >> a >> b >> c;

    cout << "Integer = " << a << endl;

    cout << "Decimal = " << b << endl;

    cout << "Character = " << c;
}
```

3. WAP to add, subtract, multiply, divide numbers.

Code:

```
#include<iostream>
using namespace std;

int main()
{
    int a, b;

    cin >> a >> b;

    cout << "Addition = " << a + b << endl;
```

```

cout << "Subtraction = " << a - b << endl;

cout << "Multiplication = " << a * b << endl;

cout << "Division = " << a / b;
}

```

4. Write a C++ program that will ask for a temperature in Celsius and display it in degree Fahrenheit. $[F=9C/5+32]$

Code:

```

#include<iostream>
using namespace std;

int main()
{
    float c, f;

    cin >> c;

    f = (9 * c / 5) + 32;

    cout << "Fahrenheit = " << f;
}

```

5. WAP to format console output using '\n', '\t', '\b', endl, setw within cout statement

Code:

```

#include<iostream>
#include<iomanip>
using namespace std;

int main()
{
    cout << "Hello\n";

    cout << "World\tC++\n";

    cout << "ABC\bD\n";

    cout << "Using endl" << endl;

    cout << setw(10) << "C++";
}

```

6. WAP to utilize assignment += and -= type operators.

Code:

```
#include<iostream>
using namespace std;

int main()
{
    int a = 10;

    a += 5;

    cout << "After += : " << a << endl;

    a -= 3;

    cout << "After -= : " << a;
}
```

7. WAP to swap two numbers using bitwise operator.

Code:

```
#include<iostream>
using namespace std;

int main()
{
    int a, b;

    cin >> a >> b;

    a = a ^ b;
    b = a ^ b;
    a = a ^ b;

    cout << "a = " << a << endl;

    cout << "b = " << b;
}
```

8. Write a program to solve the following problem

A library charges a fine for every book return late. For first 5 days the fine is 50 paise, for 6-10 days fine is one rupee and above 10 days fine is 5 rupees. If you return the book after 30 days your membership will be cancelled. WAP to accept no. of days the member is late to return the book and display the fine or appropriate message.

Code:

```
#include<iostream>
```

```

using namespace std;

int main()
{
    int days;
    float fine;

    cin >> days;

    if(days > 30)
    {
        cout << "Membership Cancelled";
    }
    else if(days <= 5)
    {
        fine = days * 0.50;

        cout << "Fine = " << fine;
    }
    else if(days <= 10)
    {
        fine = days * 1;

        cout << "Fine = " << fine;
    }
    else
    {
        fine = days * 5;

        cout << "Fine = " << fine;
    }
}

```

9. WAP to display arithmetic operations using switch-case statement.

Code:

```

#include<iostream>
using namespace std;

int main()
{
    int a, b, choice;

    cin >> a >> b;

    cout << "1.Add\n2.Subtract\n3.Multiply\n4.Divide\n";

    cin >> choice;

    switch(choice)

```

```

{
    case 1:
        cout << a + b;
        break;

    case 2:
        cout << a - b;
        break;

    case 3:
        cout << a * b;
        break;

    case 4:
        cout << a / b;
        break;

    default:
        cout << "Invalid Choice";
}
}

```

10. WAP to check whether the given number is even or odd (By using if-else and conditional operator).

Code:

```

a) #include<iostream>
using namespace std;

```

```

int main()
{
    int n;

    cin >> n;

    if(n % 2 == 0)
    {
        cout << "Even";
    }
    else
    {
        cout << "Odd";
    }
}

```

```

b) #include<iostream>
using namespace std;

```

```

int main()

```

```
{  
    int n;  
  
    cin >> n;  
  
    (n % 2 == 0) ? cout << "Even" : cout << "Odd";  
}
```